INCH-POUND

MIL-C-51540A(EA)
17 February 1994
SUPERSEDING
MIL-C-51540(EA)
21 March 1989

MILITARY SPECIFICATION

CONTROL MODULE (FOR THE M84 FILTER UNIT)

This specification is approved for use by the U.S. Army Edgewood Research Development and Engineering Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an electrical control module functioning as a distribution center for 28 volt (V) direct current (DC) and 208 V, 400 Hertz (Hz), 3-phase electrical circuits.

2. APPLICABLE DOCUMENTS

2.1 Government documents, drawings, and publications. The following Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY EDGEWOOD RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

5-19–6906 - Control module

Beneficial comments (recommendations, additions, deletions) and any pertinent data which maybe of use in irnproving this document should be addressed to: Technical Director, U.S. Army Edgewood Research, Development and Engineering Center, ATTN: SCBRD-ENE-S, Aberdeen Roving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Reposal (DD Form 1426) appearing at the end of this document or by letter,

AMSC N/A FSC 4240

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SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-19-6906 - Control module

(Copies are available from Commander, U.S. Army Information Systems Command, ATTN: SMCRI-IMO-D, Rock Island, IL 61299 -7210.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the document which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

ASTM STANDARDS

D 3951 - Commercial Packaging

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103 –1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also maybe available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-19-6906 shall conform to the specifications listed thereon and to the specific characteristics set forth on the drawings listed thereon.

3.1.2 Components. All components of the control module shall conform to the specifications and drawings listed on Drawing 5- 19–6906.

3.2 Manufacture and assembly. The control module shall be manufactured and assembled in accordance with Drawing 5-19-6906.

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3.3 Performance. The following circuits shall conform to the schematic representations on Drawings 5-19-6907 and 5-19-6908, and shall perform as specified in 3.3.1 through 3.3.7 when the applicable air pressures, voltages, and circuit sequences are applied to specific junctions on the control module and tested in accordance with 4.4.6.

3.3.1 Interconnecting circuits. There shall be continuity between the circuit points in table I and an open between the circuit points in table 11 when tested as specified in 4.4.6.2.

TABLE I. Continuity between points

Pin/connector or test point	to	Pin/connector
J 7 - D		J4-D, J7-A
J 7 - H		J4-G, J5-F, J5-B, TP-5,J4-F
T P - l		J 5 - C , J 4 - E
T P - 3		J 5 - D
T P - 4		J 5 - E

TABLE II. Open circuit between points

Pin/connector	to	Pin/connector
J 7 - A		J 7 - B , J 7 - C , J 7 - H , J 4 - A
J 7 - B		J 7 – C , J 7 - D , J 7 - H , J 4 - B
J 7 - C		J 7 - D , J 7 - H , J 4 - C ,
J 7 - D		J 7 - H
J 4 - A		J 4 - B , J 4 - C
J 4 - B		J 4 - C

3.3.2 System power circuit breaker. There shall be an open circuit between test point 2 and pin E of J4 with the system power circuit breaker open and continuity with the breaker closed when tested as specified in 4.4.6.3.

3.3.3 PE system circuit breaker. There shall be an open circuit between pins A and G of J5 and test point 2 with the protective entrance (PE) system circuit breaker open and continuity with the breaker closed when tested as specified in 4.4.6.4.

3.3.4 Power relay circuit. The power relay shall pull in and there shall be continuity between the circuit points in table III when tested as specified in 4.4.65.

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TABLE III. Continuity between points (power relay circuit)

Pin/connector	to	Pin/connector
J 7 - A		J 4 - A
J 7 - B		J 4 - B
J 7 - C		J 4 - C

3.3.5 Indicators. All indicators (see Drawings 5-19-6907 and 5- 19-6908) on the module shall light when tested as specified in 4.4.6.6.

3.3.6 Warning system. When tested as specified in 4.4.6.7, the warming system shall function as follows: The MASK indicator shall flash and the horn shall sound with a pulsating tone between 0.5 and 1.1 inches water gauge (iwg). The peak horn level shall be at least 68 dB in at least one octave band between *200* and 5000 Hz. After 15 to 25 seconds, the horn shall silence and the MASK indicator shall stop flashing and remain ON. The MASK indicator shall go out between 0.5 and 1.1 iwg.

3.3.7 DC voltage. The dc voltage shall be 30 ± 3 volts between test point 2(+) and test point 5(-) when tested as specified in 4.4.6.8.

3.3.8 Loss of power. The MASK indicator shall flash and the horn shall sound with a pulsating tone when the 208 Vac input power is lost when test as specified in 4.4.6.9.

3.4 First article. When specified (see 6.2), a sample shall be subjected to the first article inspection in accordance with 4.3.

3.5 Workmanship. The control module shall be free horn cracked, bent, or dented metal sections, sharp edges, and foreign matter (dirt, oil, or viscous materials).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contractor purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or

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supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Contractor assurance of compliance The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable ___ drawing, special packaging instruction, and specification requirements using, as a minimum, the conformance criteria specified herein.

4.13 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval. (See 6.3.) When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection. First article inspection shall apply to repair part contracts or purchase orders only.

4.3.1 Sample. The first article sample shall consist of three control modules manufactured and packaged (when military packaging is required) using the same methods, materials, equipment, and processes as will be used during regular production The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.3.2 Inspections to be performed. As determined by the Government, the first article sample items shall be examined and tested in accordance with this specification using either the first article procedures in this specification or other first article procedures which will be provided by the Government. As determined by the Government, the sample first article items maybe subjected to any or all of the examinations and tests specified in this specification and be

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inspected for compliance with any or all of the requirements of the applicable drawings and special packaging instructions.

4.33 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the items produced by one manufacturer, at one plant, from the same materials, under the same manufacturing conditions, and at the same time.

4.4.2 Sampling. Sampling shall be conducted in accordance with the classification of characteristics in 4,4.5 and, when specfied, table IV. Samples shall be selected at random.

4.43 Inspection procedure. Every item in the lot shall be inspected for critical characteristics. The lot represented shall be rejected when nonconformance to a critical characteristic is found. Sample items and, when applicable, the packaging thereof shall be examined and tested in accordance with the classification of characteristics in 4.4.5. Failure of any sample item to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented.

4.4.4 Inspection characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product or whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

4.4.5 Classification of characteristics. Quality conformance examinations and tests shall be as specified in the following classification of characteristics paragraphs. When specified herein, accept on 0 and reject on 1 attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table IV for selection of sample sizes.

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TABLE IV Sampling

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 5-19-6906
4.4.5(a)	Control module		NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
critical	None defined			
Major				
101	Performance	100 percent inspection	3.3	4.4.6
102	Components correct	Table IV, level VII	3.1.2	CE&VI
103	Soldering correct	Table IV level VII	3.2	VI
104	Electrical designations correct	Table IV, level VII	3.2	VI
105	Voltage warning sign correct	Table IV, level VII	3.2	VI
106	Workmanship	Table IV, level VII	3.5	VI

NOTES:
CE - Commercial inspection equipment
VI - Visual inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.5(b)	Packaging		SPI F5 19 6906
			NEXT HIGHER ASS

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Item completely clean and dry prior to unit packing	Table IV, level IX	5.1	VI
102	Liner bag correct	Table IV, level IX	5.1	CE & VI
103	Tape correct	Table IV, level IX	5.1	CE & VI
104	Liner bag secured as specified	Table IV, level IX	5.1	CE & VI
105	Unit pack box and closure correct	Table IV, level IX	5.1	CE & VI
106	Spacer pads correct	Table IV, level IX	5.1	CE & VI
107	Spacer pads applied as specified	Table IV, level IX	5.1	CE & VI
108	Unit pack marking correct	Table IV, level IX	5.1	CE & VI

NOTES:

CE - Commercial inspection equipment
VI - Visual inspection

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4.4.6 Performance. The control module chassis shall be connected to power supply negative potential throughout this test.

4.4.6.1 Preparation for test. Position the control module on a work bench so that the various controls can be operated during the test. Open all circuit breakers and turn the POWER switch on the control module to the OFF position. Connect the pressure tap on the control module to an incline manometer (O to 3 iwg scale, 0.02 iwg per minor increment) and a vacuum device (aspirator, mechanically actuated bellows, etc.) adjustable from O to 3 iwg. Adjust the vacuum to 2.0 iwg.

NOTE: The shielded pressure tap on the control module must be unobstructed.

Test the various circuits of the control module in the exact sequence specified in 4.4.6.2 through 4,4.6.9. Continuity is defined as less than 1-ohm resistance. Open is defined as more than 100,000-ohms resistance.

4.4.6.2 Interconnecting circuits. Use an ohmmeter to test between the circuit points specified. For all but the J7 - D to J7 - A continuity test (table I), continuity shall be defined as less than 1 ohm resistance. Continuity for the J7 - D to J7 - A circuit path shall be defined as 2 ohms or less. An open circuit is defined as more than 100,000 ohms resistance.

4.4.6.3 System power circuit breaker. Connect the ohmmeter to test point 2 and pin E of J4. The ohmmeter lead with the positive potential shall be connected to test point 2. Turn the POWER switch on the control module to the ON position. The meter shall indicate greater than 9500 ohms resistance and then continuity (less than 1.5 ohms resistance) as the SYSTEM POWER circuit breaker is depressed. Turn the POWER switch on the control module to the OFF position. “

4.4.6.4 PE system circuit breaker. Connect the ohmmeter to pin A of J5 and test point 2, then to pin G of J5 and test point 2. The meter shall indicate an open for both circuits. Depress the PE SYSTEM circuit breaker. The same two circuits shall indicate continuity when tested with the ohmmeter.

4.4.6.5 Power relay circuit With pin B of J5 at ground potential, apply 28 Vdc to pin C of J5, causing the power relay to pull in. Using the ohmmeter, test for continuity at the circuit points specified in table III for conformance to 3.3.4.

4.4.6.6 Indicators. With pin B of J5 at ground potential, apply 28 Vdc to pin A of J5. Test the “press to test” feature of all the indicators on the control module by depressing their lens covers. AU indicators shall light. Apply ground potential to pin E of J5. The LOW PRESSURE indicator shall light. Apply ground potential to pin D of J5. The OCCUPIED indicator shall light.

4.4.6.7 Warning system. Turn the POWER switch on the control module to the ON position. Slowly decrease the vacuum until the MASK indicator and horn come on. The vacuum reading

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shall be within the range specified and the warning system shall function as specified in 3.3.6. Measure the sound level of the warning horn in all octave bands from 200 to 5000 Hz with a sound meter positioned 5 feet from the born. The ground level shall be as specified in 3.3.6. The horn shall silence and the MASK indicator shall change to steady on within the time specified in 3.3.6. Slowly increase the vacuum. The MASK indicator shall go out within the range specified in 3.3.6. Continue to increase the vacuum to approximately 2.0 iwg. Depress the lens cover of the MASK indicator. The warning system shall function as specified and cease to operate when the MASK indicator is released. Turn the POWER switch on the control module to the OFF position.

4.4.6.8 DC voltage. Remove all previously applied voltages and signals. Apply a 208 Vat, 3 phase, 400 Hz wye power supply to pins A,B, and C of J7, with pin D of J7 connected to power supply neutral.

CAUTION: Connect pin H of J7 and the control module chassis to the 400 Hz power supply neutral prior to making the above power connection to ensure that the control module chassis remains at a safe electrical potential. Make sure power supply neutral is connected to earth ground.

Connect the positive lead of a suitable voltmeter (such as a voltmeter O to 50 Vdc ± 3 percent full-scale accuracy) to test point 2 and the negative lead to test point 5. The indicated voltage shall be as specified in 3.3.7.

4.4.6.9 Loss of power. Turn the POWER switch on the control module to the ON position. Remove the 208 Vac electrical power. The warning system shall function as specified in 3.3.8. Turn the POWER switch on the control module to the OFF position to silence the warning system and complete the test.

5. PACKAGING

5.1 Military packaging. For all shipments, other than interplant shipments (see 6.4), packaging (preservation, unit packing, and marking) shall be as specified in SPI P5-19-6906.

5.2 Interplant shipment When packaging is for interplant shipment (see 6.4), the collective protection equipment assembly covered by this specification shall be packaged and marked in accordance with ASTM D 3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers a control module used to provide operational status, and supply and distribute electrical power to the M84 filter unit.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see section 2)
(c) Type of packaging required
(d) Any special instructions required when acquired as a repair part
(e) First article:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.4 Interplant shipment. Packaging and marking for interplant shipment is for supplies and materials that do not directly enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or a prime contractor, between subcontractors and contractors, or from a vendor or contractor to a military arsenal, plant, or other activity for evaluation, immediate use, or further processing as specified in the applicable contract.

6.5 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Subject term (key word) listing.

airflow valve
filter unit

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A055

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-51540A(EA)	940217

3. DOCUMENT TITLE
CONTROL MODULE (FOR THE M84 FILTER UNIT)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U S Army Edgewood Research, Development and Engineering Center	(410) 671-3259	DSN 584-3259
c. ADDRESS (Include Zip Code) Tech Dir, U.S. Army ERDEC ATTN: SCBRD-ENE-S Aberdeen Proving Ground, MD 21010-5423	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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